

SLAM/VSLAM intelligent robot solution

The high-performance dual-camera robot chip

Overview

The MR813 integrates a quad-core ARM Cortex™-A53 CPU that operates at speed up to 1.5 GHz, with dual MIPI CSI and 13M ISP. It is the cost-effective solution for robot sweeper applications.

Highlights



High Performance

MR813 adopts quad-core 64-bit Cortex™-A53 CPU up to 1.5 GHz, which has powerful performance to ensure the smoothly running of various recognition algorithms like VSLAM, in-depth information extraction, and object recognition.



Dual Camera

MR813 integrates dual MIPI CSI, one ISP and H.264 encoder. With these features, MR813 supports up to two cameras (2M@60fps+2M@60fps) to meet the requirements of multiple applications such as dual TOF, one camera and one TOF, or dual camera.



Rich Peripheral

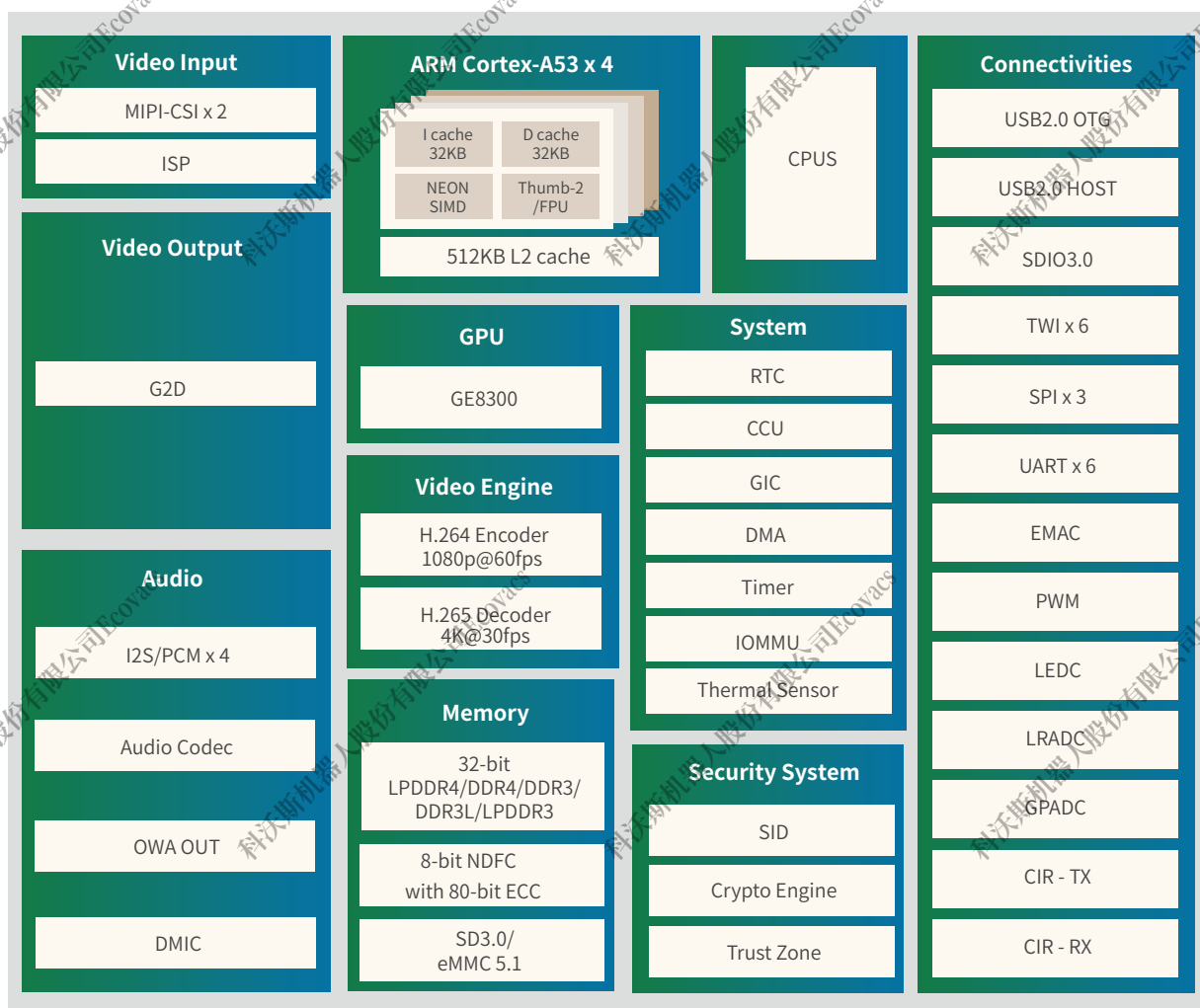
The MR813 supports rich peripheral interfaces, such as 2 x USB 2.0, 3 x SPI, 6 x TWI, 6 x UART, 5 x PWM, KEYADC, GPADC, SDIO 3.0, and so on.

Features

CPU	• Quad-core ARM Cortex™-A53@1.5 GHz • 32 KB L1 I-cache + 32 KB L1 D-cache per core • 512 KB L2 cache
GPU	• GE8300 • OpenGL ES1.1/2.0/3.2, Vulkan1.3, OpenCL1.2
Memory	• 32-bit DDR4/DDR3/DDR3L/LPDDR3/LPDDR4, support 4 GB • eMMC 5.1 • 8-bit MLC/SLC/EF NAND flash • SPI NAND flash
Video Engine	• Supports H.264 HP encoder 1080P@60fps • Supports JPEG encoder 4096 x 4096
Video Input	• Dual MIPI CSI up to 4lane and 2lane • Maximum to one camera 8M@30fps or dual camera 5M@25fps + 2M@25fps • Supports format: YUV422-8bit/10bit, YUV420-8bit, RGB888, RGB565

Others	• USB Host, USB 2.0 OTG, SDIO 3.0 • AUDIO 2 x ADC, 2 x DAC, 4 x I2S/TDM • LCD MIPI/LVDS/RGB/MCU • 3 x SPI, 6 x TWI, 6 x UART, 5 x PWM, KEYADC, GPADC
PMIC	• AXP305B
WIFI	• XR829 or others
OS	• Linux4.9/Android 10.0
Package	• LFBGA 346 balls • 12mm x 12mm size, 0.5 mm ball pitch, 0.3 mm ball size

Block Diagram



ABOUT ALLWINNER

Allwinner Technology is a leading fabless design company dedicated to smart application processor SoCs and smart analog ICs. Its product line includes multi-core application processors for smart devices and smart power management ICs used by brands worldwide.

With its focus on cutting edge UHD video processing, high performance multi-core CPU/GPU integration, and ultra-low power consumption, Allwinner Technology is a mainstream solution provider for the global tablet, internet TV, smart home device, automotive in-dash device, smart power management, and mobile connected device markets. Allwinner Technology is headquartered in Zhuhai, China.

CONTACT US

For more product info, please contact service@allwinnertech.com, or scan the QR code to follow us on Wechat.

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